Math 421/510: Homework 1

1. Define $d_1(x,y) = |\arctan(x) - \arctan(y)|$ for $x, y \in \mathbb{R}$. Show that:

   (a) d_1 is a metric on $\mathbb{R}$;

   (b) $(\mathbb{R}, d_1)$ is not complete;

   (c) d_1 produces the same topology (collection of open sets) on $\mathbb{R}$ as the usual metric $d(x,y) = |x-y|$.

2. Consider the following spaces of sequences:

$$\ell^\infty = \{x = (x_1, x_2, \dots) \mid x_j \in \mathbb{C},\ \|x\|_\infty = \sup_j |x_j| < \infty\} \qquad \text{(bounded sequences)}$$

$$\ell_0 = \{x = (x_1, x_2, \dots) \mid x_j \in \mathbb{C},\ \lim_{j\to\infty} x_j = 0\} \qquad \text{(sequences tending to 0)}$$

$$\ell_c = \{x = (x_1, x_2, \dots, x_m, 0, 0, \dots) \mid x_j \in \mathbb{C},\ m \in \mathbb{N}\} \qquad \text{(finite sequences).}$$

   Note that: $\ell_c \subset \ell_0 \subset \ell^\infty$, and all are vector spaces.

   (a) show that $\|\cdot\|_\infty$ is a norm on ℓ^∞ (hence also on ℓ_c and ℓ_0);

   (b) in class we showed that $(\ell^\infty, \|\cdot\|_\infty)$ is complete (hence Banach). Show that ℓ_c is not complete in the metric produced by $\|\cdot\|_\infty$;

   (c) show that $(\ell_0, \|\cdot\|_\infty)$ is complete (hence Banach);

   (d) show that ℓ_0 is the closure of ℓ_c (the smallest closed set which contains it) in $(\ell^\infty, \|\cdot\|_\infty)$.

3. Show that on $C^1([0,1])$:

   (a) $\|f\|_0 = |f(0)| + \|f'\|_\infty$ is a norm (where, recall $\|g\|_\infty = \sup_{x\in[0,1]} |g(x)|$);

   (b) $\|\cdot\|_0$ is equivalent to the norm $\|f\|_{C^1} = \|f\|_\infty + \|f'\|_\infty$.

4. Let X, Y, Z be normed vector spaces. Show:

   (a) (leftover from class) the operator norm is a norm on $L(X,Y)$;

   (b) if $T \in L(X,Y)$, $S \in L(Y,Z)$, composition $ST \in L(X,Z)$ with $\|ST\| \le \|S\|\,\|T\|$;

   (c) for $X = C([0,1])$, $\|\cdot\|_\infty$, find $S, T \in L(X,X)$ with $\|ST\| < \|S\|\,\|T\|$.

5. Consider the following spaces of sequences:

$$\ell^1 = \{x = (x_1, x_2, \dots) \mid x_j \in \mathbb{C},\ \|x\|_1 = \sum_{j=1}^\infty |x_j| < \infty\} \qquad \text{(absolutely summable sequences)}$$

$$\ell_0 = \{x = (x_1, x_2, \dots) \mid x_j \in \mathbb{C},\ \lim_{j\to\infty} x_j = 0\} \qquad \text{(sequences tending to 0)}$$

   (both are vector spaces). Show that

   (a) $\|\cdot\|_1$ is a norm on ℓ^1;

(b) for $y = (y_1, y_2, \dots) \in \ell^1$, the map $f_y : \ell_0 \to \mathbb{C}$ given by $x = (x_1, x_2, \dots) \mapsto y \cdot x = \sum_{j=1}^{\infty} y_j x_j$ is a bounded linear functional on ℓ_0, $\|\cdot\|_\infty$ (i.e. $f_y \in (\ell_0)^*$);

(c) $\|f_y\| = \|y\|_1$;

(d) if $f \in (\ell_0)^*$, then $f = f_y$ for some $y \in \ell^1$;

(e) the map $y \mapsto f_y$ is an isometric isomorphism $\ell^1 \to (\ell_0)^*$.

6. On $(\ell^\infty, \|\cdot\|_\infty)$, define the shift operator $S : (x_1, x_2, \dots) \mapsto (x_2, x_3, \dots)$, the subset $Y = \{x - Sx \mid x \in \ell^\infty\} \subset \ell^\infty$, and the sequence $z = (1, 1, 1, \dots) \in \ell^\infty$. Show:

   (a) S is linear and Y is a subspace;

   (b) $\operatorname{dist}(z, Y) = \inf_{y \in Y} \|z - y\|_\infty = 1$;

   (c) by applying (a Corollary of) the Hahn–Banach theorem to $\overline{Y}$ and z, that there is $T \in (\ell^\infty)^*$ with $T\restriction_Y = 0$, $T(z) = 1$ and $\|T\| = 1$;

   (d) T is shift-invariant: $TS = T$;

   (e) if $x = (x_1, x_2, \dots) \in \ell^\infty$ satisfies $\lim_{k\to\infty} x_k = L$ then $Tx = L$.